

# Create Your Own Linux Distro

We bet you didn't think it was possible!

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Wouldn't it be ultra-cool to carry around your very own operating system with you? Or share an OS with friends? Have you wanted to build an OS according to your requirements and needs? If you've answered "yes" to all the above, you're in for a treat! Here, we show you how to build your very own, customised Linux distribution (distro).

For creating our distro, we used Ubuntu 7.04 as a base and built over it. We wanted an OS that would allow us to do the usual Internet browsing, creating documents and spreadsheets, e-mailing, and playing movies and videos. Here's how to do it.

**STEP 1** Get hold of the Ubuntu 7.04 ISO file from last month's (Anniversary) issue and install it. You'll need to be running in Ubuntu while you create the distro.

**STEP 2** Make a list of all the essential programs you need. We chose NTFS Read/Write Support, The GIMP, Gaim, Automatrix2, Flash 9 player with the Mozilla plugin, W32 Codecs, Audio / Video codecs (Bad and Ugly versions), and VLC Media Player to make the distro complete.

**STEP 3** We now need a program to do our dirty work because, quite frankly, building a Live CD the Linux From Scratch (LFS) way can be quite a pain in the a\*\*\*. We chose Reconstructor, a Linux CD creator. It's a small download, from <http://reconstructor.aperantis.com/>

**STEP 4** Fire up Reconstructor with "sudo python reconstructor.py" in your Linux Terminal, and you'll be told which dependencies you need to install before you run Reconstructor.

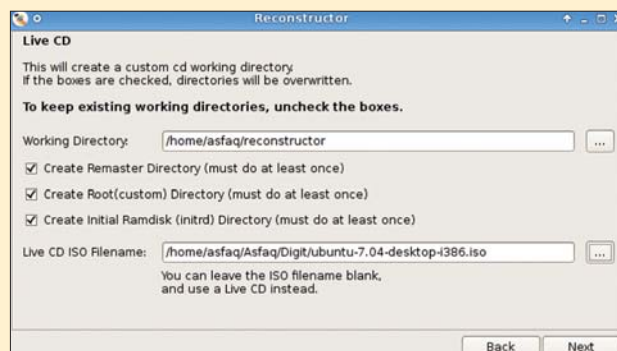
**STEP 5** Go to System > Administration > Synaptic Package Manager > Settings > Repositories to make sure the Universe and Multiverse repositories are selected. Open a terminal and run "sudo apt-get install squashfs-tools gcc rsync libbogl-dev libusplash-dev gpg dpkg-dev fakeroot apt-utils" in the terminal window. You need to be online for this to work.

**STEP 6** After downloading the dependencies, start Reconstructor again and you'll be greeted by a welcome screen.



Reconstructor's welcome screen

**STEP 7** In this new screen, the locations of working directories need to be set up—they tell the program where all the files are stored locally. Select the "Create Remaster", "Create Root", and "Create Initial Ramdisk" options. These options are used to create the cdrom directory tree, the root filesystem directory tree, and the initial ramdisk tree.



Make sure all the boxes are checked

**STEP 8** The last option in the window asks for the location of the main Ubuntu ISO file, which will be used to replicate the directory structure. Click the "..." button next to the Live CD ISO Filename option, and navigate to where the Ubuntu ISO file is located; then click Next.

**STEP 9** This screen is divided into various tabs, the first being "Boot Screens". As the name suggests, this is where you change the first screen that is shown (the splash screen) and also the image shown after you select the boot options. To select an image here, click the "Generate" button. Select the source image (which has to be a 256-colour PNG) and choose where to save it. Reconstructor will copy